

DASC7606

Deep Learning Preliminary

Dr. Bethany Chan 陳敏兒

Professor Francis Chin 錢玉麟

2026

Why take Deep Learning?

Ask the following questions

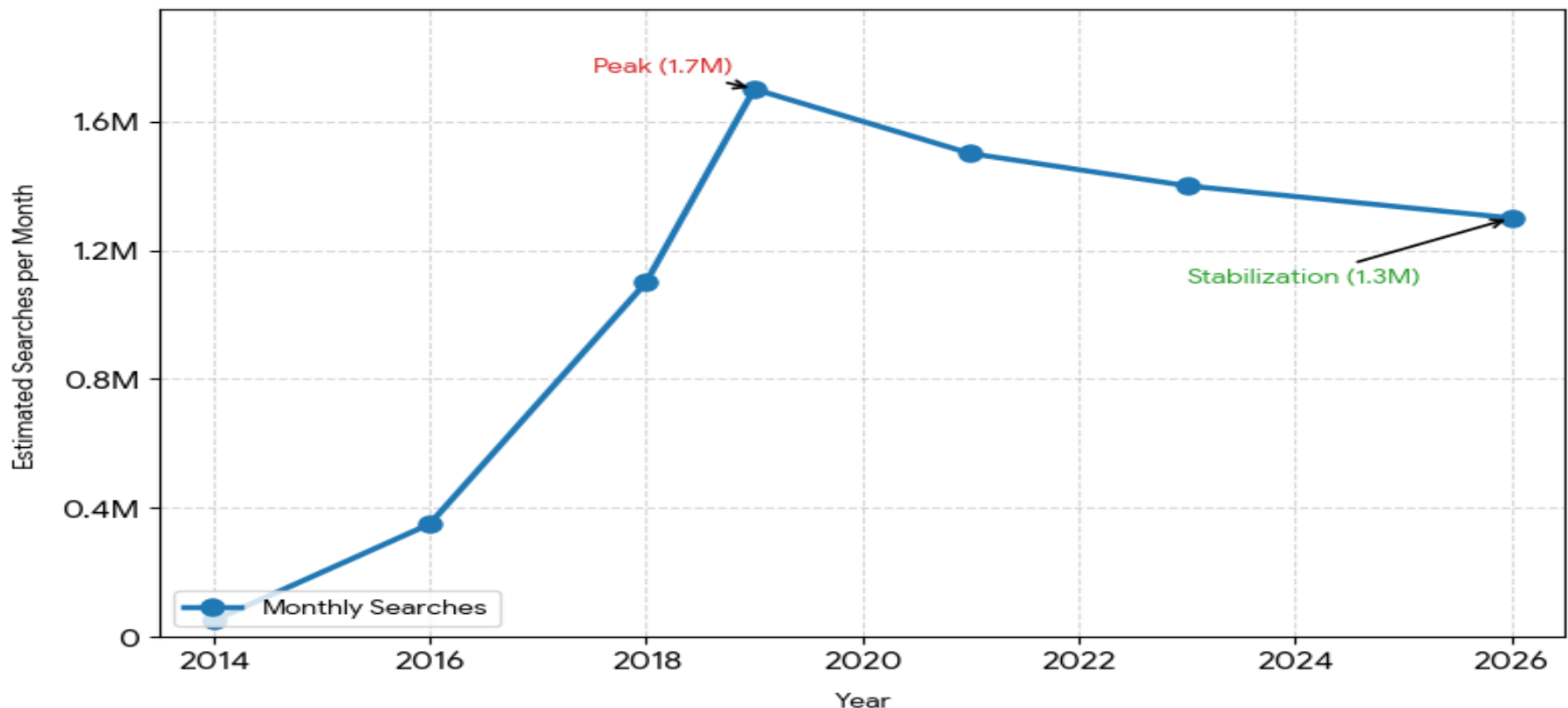
- Has the interest in AI actually increased in recent years?
- What kinds of AI jobs are trending?
- What are the most in-demand skills for these jobs?

How to measure the general interest in “Deep Learning”



No. of searches about “Deep Learning” on Google indicates its popularity.

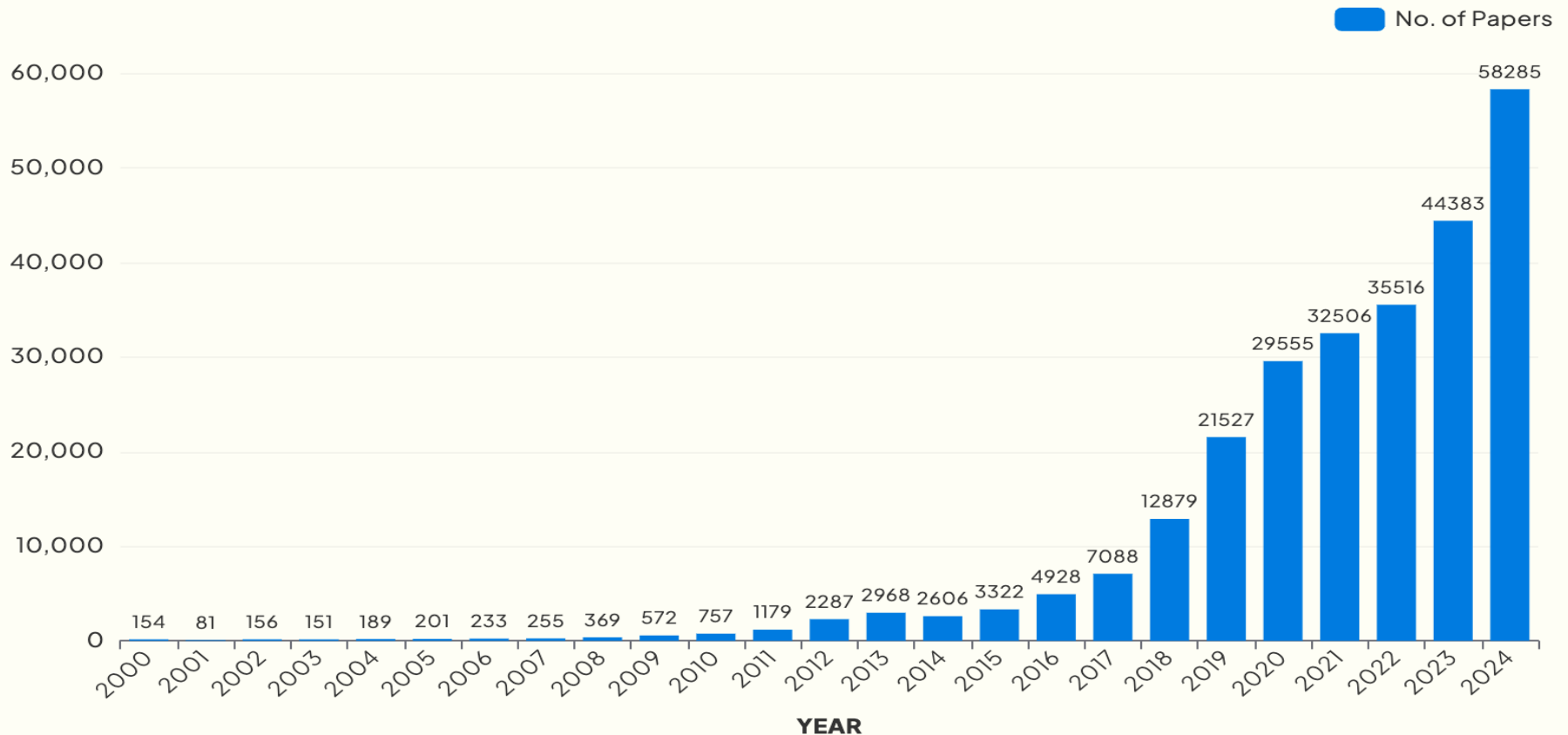
Estimated Global Monthly Google Searches for "Deep Learning" (2014–2026)



Interest in Deep Learning continues to increase – Research Community



Number of AI-related research papers published, from 2000–2024

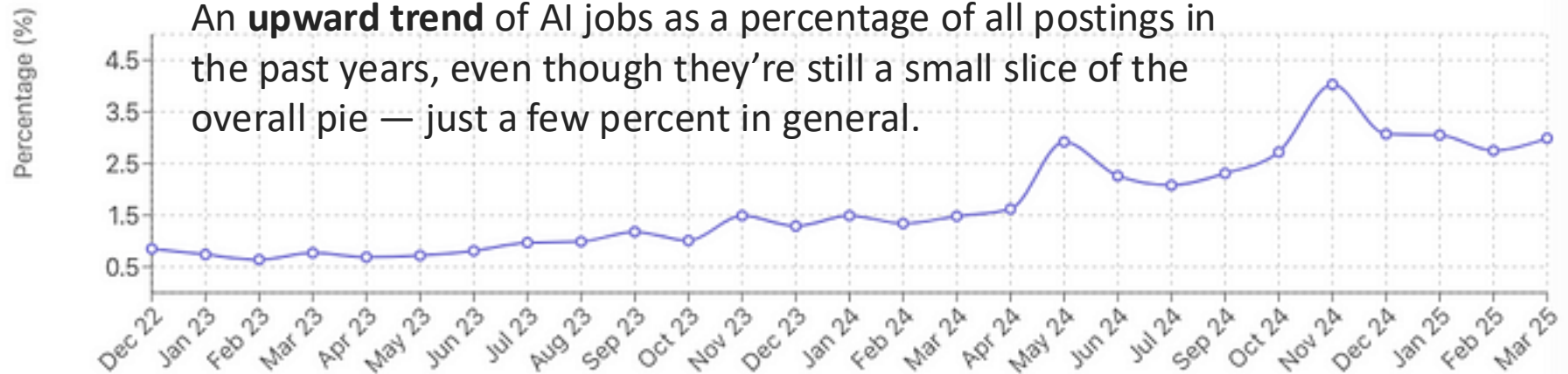


Deep Learning research effort continues to grow

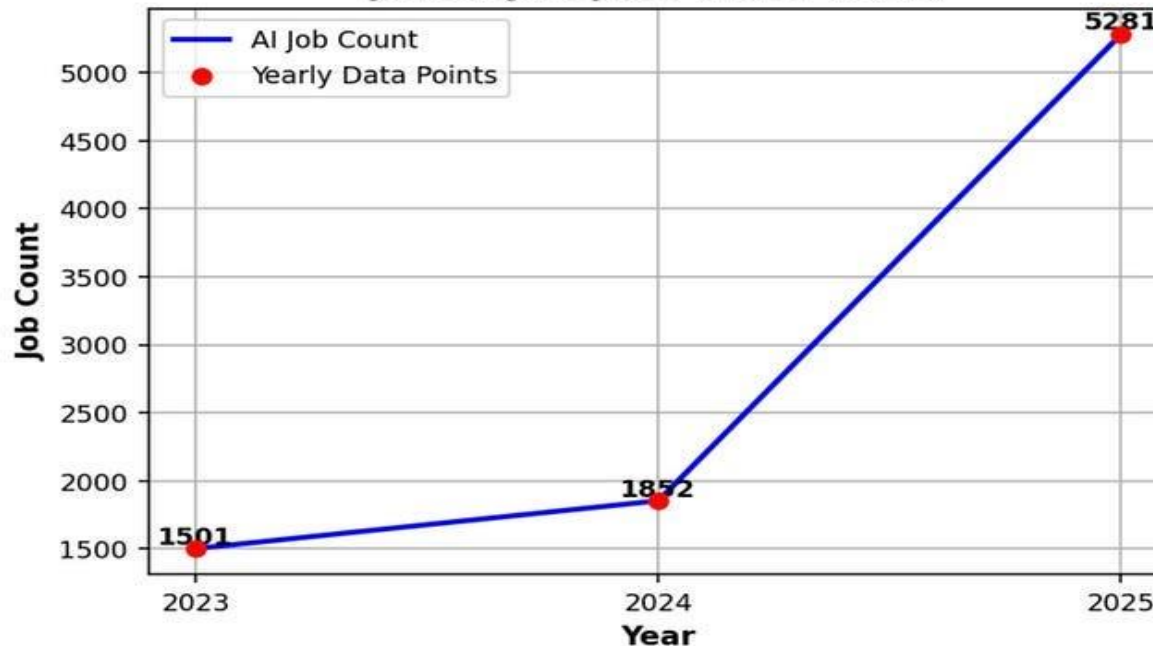
Are AI-related job postings increasing?

<https://medium.com/@vuthihienthu.ueb/should-data-scientists-pivot-to-ai-fb09d4836e8a>

Percentage of AI Jobs Out of All Data-Related Jobs



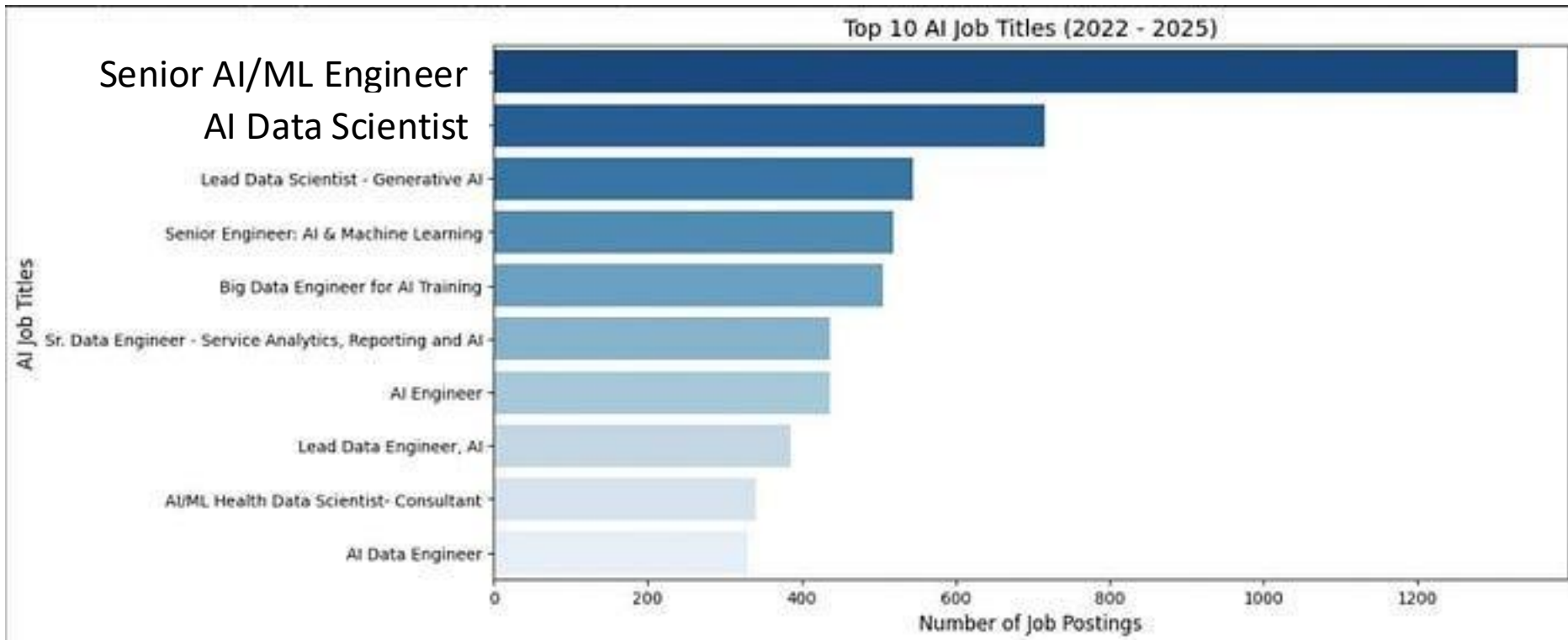
January AI Jobs (2023-2025)



A significant jump in AI job postings just one year.

About 5,000 in January 2025 compared to less than 2,000 in Jan 2024.

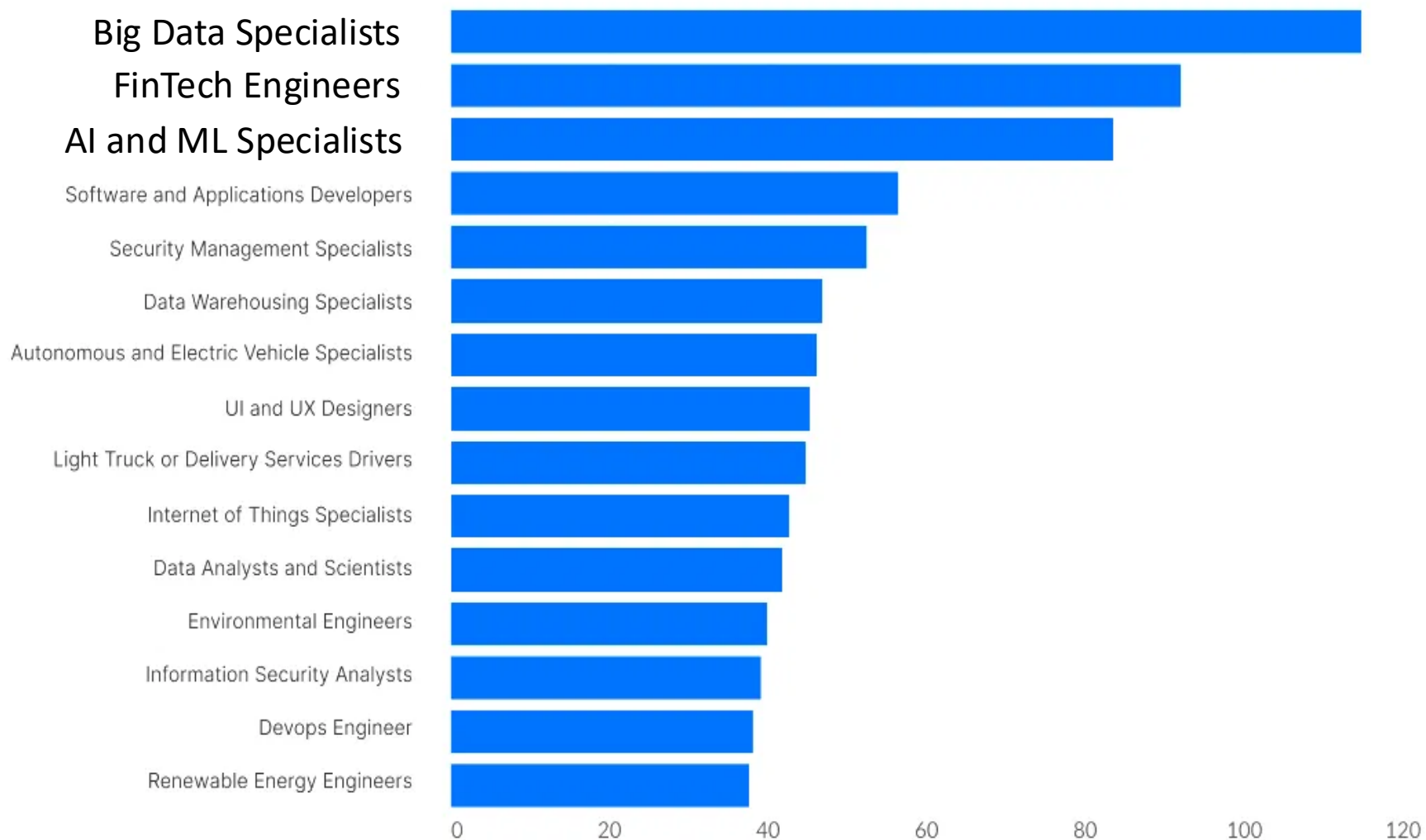
What Are These AI Jobs, Exactly?



What exactly does an AI Engineer do?

- AI Engineers lean towards the product and user end.
- They **DON'T BUILD** the AI models, but rather build applications that make use of pre-trained AI models to solve a specialized business problem.

Top fastest growing jobs, 2025-2030



Source: World Economic Forum, Future of Jobs Report 2025



GLOBAL ADMISSION

What Are The Required Skills?

Three key layers of the AI engineering stack:

1. **Application Development** where you craft *effective prompts*, incorporating *relevant context* (like a knowledge base), potentially also building an *agent workflow*, and designing a solid *user interface*.
2. **Model Development** where you fine-tune models, optimizing inference, and handling data collection and engineering.
3. **Infrastructure layer** where you manage cloud resources and compute power, monitor and maintain AI applications at scale.

Skill/knowledge needed: Python, Pytorch, Azure, AWS

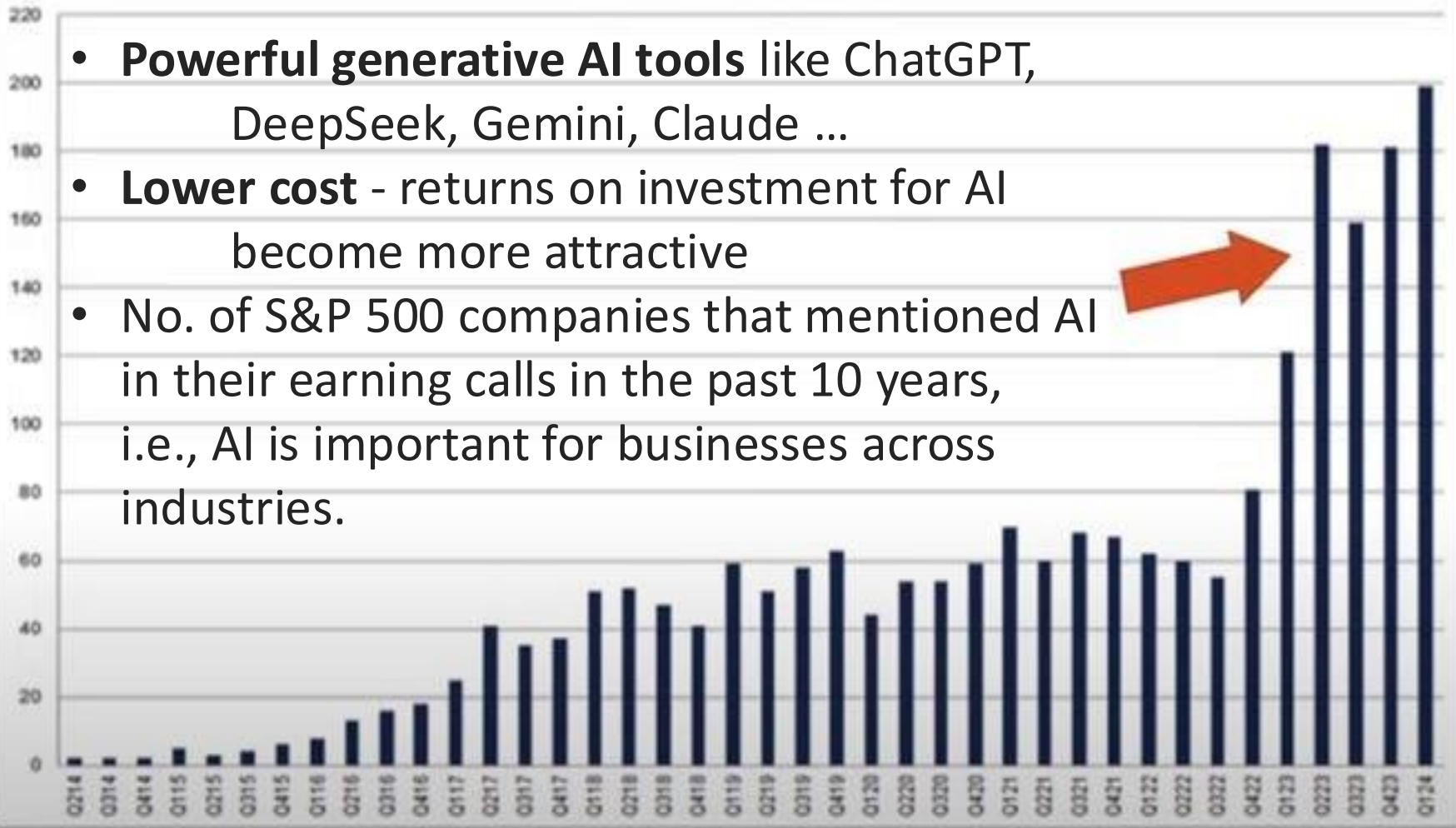
Course objectives: fundamentals and basic concepts of Deep Learning (not only on LLM for Application and Model development).

So... Why is This Happening?



of S&P 500 Cos. Citing "AI" on Earnings Calls: 10-Year
(Source: FactSet)

- **Powerful generative AI tools** like ChatGPT, DeepSeek, Gemini, Claude ...
- **Lower cost** - returns on investment for AI become more attractive
- No. of S&P 500 companies that mentioned AI in their earning calls in the past 10 years, i.e., AI is important for businesses across industries.



AI bubble

- OpenAI CEO Sam Altman has reportedly said that he believes **AI could be in a bubble**, comparing market conditions to those of the dotcom boom in the 1990s.
- No AI company or AI infrastructure is profitable.
They are all losing money
(OpenAI's annual recurring revenue is on track to pass \$20 billion this year, but that despite that, it remains unprofitable)
- AI the most important thing to happen in a very long time, but investors as a whole are overexcited about AI.
- Alibaba co-founder Joe Tsai, Bridgewater Associates' Ray Dalio and Apollo Global Management chief economist Torsten Slok have all raised similar warnings

MIT report: 95% of generative AI pilots at companies are failing



BY **SHERYL ESTRADA**

SENIOR WRITER AND AUTHOR OF CFO DAILY

August 18, 2025 at 6:54 AM EDT

More than half of generative AI budgets are devoted to sales and marketing tools, yet MIT found the biggest ROI in back-office automation

2026 Business Reality: Embrace AI or Fade Away

**“Companies have to race to build AI or they will be made uncompetitive. Essentially, if your competitor is racing to build AI, they will crush you.”
— Elon Musk**

Elon Musk made this statement in 2017 during a discussion on the risks and competitive pressures of AI development.

5 Industries AI Will Completely Take Over by 2026

<https://pub.aimind.so/5-industries-ai-will-completely-takeover-by-2026-d16d36b70cce>

Goldman Sachs warns that AI could displace 300M jobs worldwide (not downsize/restructured, **replaced**), 76,440 jobs in 2025 alone.

- 1. Legal & Compliance** - *When Robots Know The Law Better Than Lawyers*
Reality: A lawyer can review 20 contracts/day. AI can process **2,000** with **99.7% accuracy**. (contract reviewer, paralegal researcher, compliance officer, document analyst, legal secretary)
- 2. Accounting & Finance** – *no coffee breaks, no calculation errors, work 24/7.*
Reality: JPMorgan's COIN (Contract Intelligence) program processes in seconds what used to take legal/accounting teams 360,000 hrs annually. (account payable/receivable, tax preparation/filing, audit procedure, financial reporting/analysis,, budget forecasting, expense management)
- 3. Healthcare Administrator** – *medical claims, reduce process time to minutes*
(medical billers, Insurance claim processors, Patient registration specialist, Medical records technician, authorization coordinator, data entry operator)
- 4. Customer Support & Call Centres** – *more patient and knowledgeable*
handle 80% of routine customer inquiries without human intervention
- 5. Content & Data Management** – write articles, analyse data, and generate reports faster (content writer, copywriter, market researcher, transcriptionists social media manager, administrative assistant)

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Deep Learning

Course Learning Outcomes

- Able to **understand deep learning concepts**.
- Able to **understand the application** of **deep learning concepts** to various problems.

Topics Coverage

- **Perspectives:**
history of AI, basic concepts, neural network, training
- **Vision:** CNN, AlexNet to Resnet, applications and concepts
- **Generative AI:** DeepDream, VAE, GAN, Dalle, Diffusion
- **Natural Language Processing:** word embedding, RNN/LSTM, ELMo, Transformer, BERT, LLMs, GPT, ChatGPT, DeepSeek...
- **Other topics:** Deep Reinforcement Learning, Ethics

Particulars about DASC7606 ...

Prerequisites

- Basic programming (Python)
- No prior knowledge in artificial intelligence or machine learning.
- Knowledge of **algorithms**, **calculus**, and **linear algebra** would be an advantage

Assessment

Assignments – 2 programming assignments (50%)

Examination – 2 hours (50%)

GPU farm and Python programming self-learn
(TA will give guidance to learn)

Others

Lectures will NOT be recorded, ppts posted after lecture

Breaks: 10 minutes + early end

Timetable

Lecture	Friday		Tentative
	(10:00-13:00) LG1/F KKL	(19:00-22:00) LE -1	
1	September 4, 2026		
2	September 11, 2026 Sept 12, 2026 14:00-17:00 (KB223, 2/F Knowles)		
3	September 18, 2026		Issue Assignment #1
4	September 25, 2026		
5	October 2, 2026		
	Reading Week		Issue Assignment #2
6	October 30, 2026		
7	November 6, 2026		
8	November 13, 2026		
9	November 20, 2026		
10	November 27, 2026		
Exam	December 3-23 (date unknown)		17

Teaching Assistants

Solely responsible for the assignments

- **Dunjie Lu** dunjielu1219@connect.hku.hk
- **Haomin Bao** baohaomin@connect.hku.hk
- **Xiangrui Yang** xiangruiyang@connect.hku.hk

They will handle **ALL** matters
(support, questions, submission, grading, ...)
related to assignments.

They can be contacted through Moodle, email or
any other means.